UNIVERSITY OF LEEDS

Module Handbook

OCOM5205M Robotics (H02) 2023/24

Contents

Module Handbook 1
Contents 1
Accessibility 2
Welcome to the Module 2
Module Team 2
Module Objectives 3
Learning Outcomes 3
Syllabus 4
Unit 1 – K-Armed Bandit and Markov Decision Processes (MDP) 4
Unit 2 – Dynamic Programming and Monte Carlo Algorithms 4
Unit 3 – Temporal Difference: Prediction, Control and Planning (Tabular) 5
Unit 4 – Temporal Difference: Prediction, Control and Eligibility (Linear Function Approximation) 5
Unit 5 – Temporal Difference: Prediction, Control and Eligibility (Non-Linear Function Approximation) 6
Unit 6 – RL Applied on Other Domains - Optional 6
Key Dates 6
Module Assessment 8
Assessment 1: Multiple choice questions test 8
Assessment 2: Robotics and RL Project: Reinforcement Learning on Robot Navigation 12

© University of Leeds

Reading List 18

Additional Information and Support 19

Key Documents 19

Skills@Library 19

General Module Forum 19

Student Success Advisor 19

Recording Policy 19

Complaints 20

Accessibility

For information in alternative formats please email DESstudents@leeds.ac.uk or speak with your Student Success Advisor.

Welcome to the Module

Welcome to OCOM5205M Robotics.

This handbook contains administrative information related to the module to help you to organise study time for the module. We hope that you find this material useful and the work you do on the module rewarding.

Module Team

Your module leader is Abdulrahman Altahhan. You will have the opportunity to speak to him during the live weekly webinar sessions. If you need to contact Abdulrahman directly, you can find his details in the ‘Staff Information’ section of Minerva.

Module Objectives

This module aims to introduce you to the principles of reinforcement learning and its application on intelligent robotics. You will develop your knowledge of the concepts and techniques of reinforcement learning as a major paradigm of machine learning that is used successfully to allow a robot (physical and virtual) to learn or teach itself to perform a task. You will cover traditional applications of robotic walking (robot learns to walk), navigation (e.g. a robot navigating from one position to another) and manipulation of objects (e.g. a robot moving its arm to reach an object). Activities support your application of these skills, before you go beyond traditional robotics tasks to the wider realms of intelligent agents and of deep reinforcement learning, including gaming (Atari, Alpha Go etc.) We also introduce you to the software tools used in modern reinforcement learning and robotics to develop resilient intelligent agents.

Learning Outcomes

Successful completion of this module will help you to:

- Apply the fundamental concepts in modern robotics, including robot navigation and manipulation, to typical scenarios where we use robots
- Demonstrate the fundamental concepts of reinforcement learning techniques for some common tasks completed by robots
- Critique the challenges of training an intelligent agent
- Select and develop the most appropriate software to control intelligent agents

© University of Leeds

Syllabus

Unit 1 – K-Armed Bandit and Markov Decision Processes (MDP)

Unit 1 Learning outcomes:

1. understand the armed-bandit problem and its simplification of isolating the action space
2. understand the value function and the action-value function and their essential roles in RL
3. understand the difference between associative and non-associative problems
4. understand the underlying theory of RL including MDP and Bellman equation
5. understand the difference between prediction and control in RL settings

Unit 2 – Dynamic Programming and Monte Carlo Algorithms

Unit 2 Learning outcomes:

1. understand how to predict the value function for a policy in a tabular settings
2. understand how to control an agent by inferring its policy from an action-value function
3. understand the idea of generalised policy iteration (GPI) that is utilised by many RL methods
4. understand the difference between full-backup action-value based control methods and direct policy estimation control methods
5. understand how MC obtains an unbiased but high variance estimation via interaction with using the environment
6. understand how REINFORCE obtains an unbiased but high variance estimation via interaction with using the environment

© University of Leeds

Unit 3 – Temporal Difference: Prediction, Control and Planning (Tabular)

Unit 3 Learning outcomes:

1. appreciate the importance of bootstrapping and its essential role in RL
2. understand n-steps methods and the trade-offs that n represents
3. appreciate the different tabular methods and their application
4. understand the difference between n-steps-backup action-value based control methods and direct policy estimation control methods
5. understand how TD obtains a biased but low variance estimation via interaction with using the environment
6. understand how actor-critic obtains a biased but low variance estimation via interaction with using the environment
7. understand on-line and off-line algorithms trade-offs
8. understand planning methods and how to build a model along the way by learning

Unit 4 – Temporal Difference: Prediction, Control and Eligibility (Linear Function Approximation)

1. apply RL to control an agent in a more complex environment representation
2. understand on-policy and off-policy algorithms trade-offs
3. understand the theoretical guarantees of RL algorithms in tabular settings and their practical limitations
4. understand the theoretical guarantees of RL algorithms with function approximation and the deadly triad

© University of Leeds

Unit 5 – Temporal Difference: Prediction, Control and Eligibility (Non-Linear Function Approximation)

Unit 5 Learning outcomes

1. understand how to predict the value function for a policy using function approximation
2. understand eligibility traces methods and the trade-offs that their depth represents
3. understand how to control an agent by inferring its policy from an action-value function with function approximation

Unit 6 – RL Applied on Other Domains - Optional

Unit 6 Learning outcomes

1. Study more advanced RL algorithms such as TRPO and PPO
2. apply RL to control an agent in Games
3. apply RL to control a robot
4. apply RL to control an agent in a simple environment

Key Dates

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	**4th March Teaching starts**	**5th**	**6th**	**7th**	**8th**
Week 2	**11th** **Unit 1 webinar** **16:00 – 17:00 UTC (London)**	**12th**	**13th**	**14th**	**15th**

© University of Leeds

Week 3	**18th** **Unit 2 webinar** **16:00 – 17:00 UTC (London)**	**19th**	**20th**	**21st**	**22nd**
Week 4	**25th** **Unit 3 webinar** **16:00 – 17:00 UTC (London)** **Assessment 1 release**	**26th**	**27th**	**28th March** **University Closed Easter**	**29th March** **University Closed Easter**
Week 4 (cont'd)	**1st April** **University Closed Easter**	**2nd April** **University Closed Easter**	**3rd**	**4th**	**5th**
Week 5	**8th** **Unit 4 webinar** **16:00 – 17:00 UTC+1 (London)** **Assessment 1 deadline**	**9th**	**10th**	**11th**	**12th**
Week 6	**15th** **Unit 5 webinar** **16:00 – 17:00 UTC+1 (London)**	**16th**	**17th**	**18th**	**19th**
Week 7	**22nd** **Unit 6 extended* webinar** **16:00 – 17:30 UTC+1 (London)**	**23rd**	**24th**	**25th**	**26th**

© University of Leeds

Study break	**29th** **Assessment 2 deadline**	**30th**	**1st May**	**2nd**	**3rd**

Please see the module area in Minerva for specific timings for webinars and assessment submission deadlines.

Please note that **the University of Leeds will be closed** in the following dates:

- Easter holiday from Thursday 28th March and will reopen on Wednesday 3rd April
- Daylight saving time starts on Sunday 31st March 3 AM.

Module Assessment

You should familiarise yourself with the Digital Education Service Code of Practice on Assessment (CoPA) before you start your first assignment. The CoPA contains important information about marking, late submissions, resits, mitigating circumstances and more.

Assessment 1: Multiple choice questions test

Summary

20 multiple choice questions

Deadline

Released: Monday 25th March at 08:30 UTC (London)

Due: Monday 8th April 2023 at 08:30 UTC +1 (London)

Purpose

The purpose of this assignment is to help you to evaluate your understanding of robot localisation and imitation learning covered in Units 1 to 3.

Task

You will have 100 minutes to complete a series of multiple choice questions based on the content of Units 1 to 3. This will allow you to practice and apply your learning from the module. Each question title identifies which Unit it relates to.

The assessment will be completed in Gradescope. You should record all your answers in Gradescope.

There are 20 questions in total, and they vary in the total number of points allocated for your answer.

Accessibility

Information for students about the accessibility of the material in this multiple choice question assessment:

- This assessment contains some jpeg images, which do not meet accessibility standards. We have written simple alt text for these images, which does not fully describe their complexity.
- If you are unable to access any of the content of this module for disability related reasons, please contact your module leader in the first instance. You should also ensure that you have registered for support from Disability Services.

Academic integrity

As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' guidance on Academic Integrity. Please note that by submitting this piece of work you are agreeing to the University's Declaration of Academic Integrity.

© University of Leeds

You will need

Before you start the test, you might find it useful to have ready:

- a pen and paper
- a calculator

Submission

Complete this exercise on Gradescope via the link in the Submit my Work module area.

- The link to access the test will be available in the Submit my Work folder once the test window is open.
- You will have one attempt at the test.
- You have a time limit of **100 minutes** to complete the test once you have begun.
- **Please note:** Once you start the test the timer **cannot be stopped**. It will continue to countdown the time even if you close the test or navigate away from Gradescope.
- You will not be able to complete the test after the deadline has passed. If you start the test less than 100 minutes before the deadline you will not receive the full-time allocation.
- You may change any of your answers until the end of your time limit or the deadline (whichever comes first).
- Your answers for this assignment will be automatically saved as you go through the test.
- Once you have completed all your answers you will need to click on submit button.
- Mistakes and problems with submission cannot normally be rectified once the time limit has expired or the deadline has passed.

© University of Leeds

Generative AI Category

Generative AI category: Red

This means: you are **not** permitted to use AI tools for this assessment.

Whilst AI tools cannot be used in this assessment, you are still permitted to use dictionaries, thesauri, spelling and grammar-checking software to help identify and correct spelling mistakes and grammatical errors (even if they are powered by Gen AI). However, you should not use any software to rewrite sentences or make substantive changes to your original text.

Marking

This is a compulsory assessment for OCOM5205M. It is worth **30%** of your overall module grade. There are 20 questions, with a total of 50 marks available.

Feedback

Grading for this assessment is automatic. In line with normal practice for exams, only the overall mark for the test will be available on Gradebook once marking is complete.

© University of Leeds

Assessment 2: Robotics and RL Project: Reinforcement Learning on Robot Navigation

Summary

This assessment covers your learning from Units 1 to 6. In this project, you will create a robot controller that allows a robot to learn how to navigate in an indoor environment to achieve a simple task. You will be given an indoor environment model with a launch file to launch the environment with a turtlebot3 robot. The task is to go from the start position to either of the pillar areas without hitting obstacles. You can use the odometry and the laser scan sensor to feed into the learning model.

Deadline

Due: Monday 29th April 2024, 8:30 UTC+1 (London)

Task

You need to create three trainable robot navigation models for this task. Your robot models will navigate in an indoor environment without hitting the walls to reach a goal destination near one of the pillars in the environment.

Robot model and environment specification

You will use a burger TurtleBot 3 model in Gazebo. You should follow the instructions in the link to install all required turtlebot3 packages according to your ROS (it is already set up for your Azure account). To make training of RL solution more efficient, you must adjust the number of laser distance sensor (LDS) from 360 to a more manageable count, such as 24.

You are provided with the environment files which you can download here. You must create and build a new package called turtlebot3_assessment in your src in your VM. You can then launch the assessment environment in the terminal.

We have also provided you with another simpler environment to practice if needed, your assessment must use the environment accessed via turtlebot3_assessment2.launch. Your

Jupyter notebook need to be placed in the src of the assessment package. You will need to copy all the provided notebook lessons and place them inside the same folder of your assessment notebook to access the code.

Instructions for training the navigational models of the robot

You can feed the navigational models the robot's odometry and or the scan sensor reading of the robot's location (see below which one to use in each model). The robot must depend entirely on the navigational models to navigate the environment. Reaching the goal vicinity within 2 feet of a pillar without hitting an obstacle is a success and marks the end of an episode. Hitting an obstacle reset the robot's position and results in a negative reward. You will terminate the episode using the robot's and the pillar's locations. The robot needs to be left to learn by itself. You should design a suitable reward to aid the robot's learning.

The sequence of steps for this task

1. You will use a Jupyter notebook to provide your solutions. You would need to break the task into stages, all of them with the same above objective:
2. Create an Env class that isolates the ROS publish-subscribe logic and deals with the environment from the RL solutions. You may use some of these examples to develop your class. The Env class will be the base for your solutions. You may inherit it and change some of its functionality for each individual model as you see fit.
3. Tabular method for autonomous navigation:
4. Use the robot's odometry (position and orientation) for state representation
5. Divide the environment into a grid.
6. Create a suitable reward function.
7. Train your model using one of the tabular control methods we covered in our lessons
8. You must use the provided code in the lessons and not write your RL from scratch
9. Justify your choice of the algorithm and justify the hyperparameters you set during learning.

© University of Leeds

10. Vectorised linear-based RL model for autonomous navigation:
11. Use the robot's scan and the odometry for state representation.
12. Create a suitable reward function.
13. Train your model using one of the function approximation control methods we covered in our lessons.
14. You must use the provided code in the lessons and not write your RL from scratch.
15. Neural network-based RL model for autonomous navigation:
16. Use a shallow neural network and an experience replay buffer to train a model similar to DQN and DDQN.
17. Use the scan and the odometry as observation input for the network.
18. Create a suitable reward function.
19. Train using one of the function approximation control methods we covered in our lessons.
20. You must use the provided code in the lessons and not write your RL from scratch, but you can use any neural network architecture that you see fit.

You are free to use either 3 actions or 5 actions in all of the above models. The robot starts around the middle of the environment but not exactly. You must not change the starting position or orientation of the robot in the given environment. You must show in each stage the accumulated reward and number of steps plots for each episode. You can use suitable demoXX() functions we provided or directly set the corresponding visualisation parameters. You can add other plots of your choice as well. You must report how long it took you to train each model. You can train as many episodes as you find suitable/practical. You should list the hardware specs used for training, particularly whether you used GPUs and their speed and memory.

© University of Leeds

Presentation

Your report should adhere to the guidelines on presentation of coursework as set out in **Section 3.2a of the Code of Practice on Assessment.**

Referencing

It is important that you show that you have engaged with appropriate literature and make efforts to explicitly cite all the sources that you have used or have shaped your thinking.

Academic integrity

As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' guidance on Academic Integrity. Please note that by submitting this piece of work you are agreeing to the University's Declaration of Academic Integrity.

Generative AI Category

Generative AI category: Amber

This means: AI tools can be used in an assistive role for specific designed processes as specified by the module tutor.

Submission

Coursework should be submitted via the link in the Submit my Work area.

© University of Leeds

This is a compulsory assessment for OCOM5205M Robotics. **It is worth 70% of your overall module grade.** Your technical report will be marked out of a total of 100 marks, according to the mark scheme given below. You will need to write a technical report with the same sections as the marking scheme; the word count should be around 2000 words. The report must explain the important details of your project and include the learning plots and the length of training execution time as well as your neural network architecture and the hyperparameters settings for it and the RL algorithms with justification. You will be marked based on the quality of the technical report.

Files to submit on Minerva:

1. The project's technical report that you will need to submit is in the form of a Jupyter notebook that includes your code and explanations. This notebook must utilise the code we provided in the lessons and not create a new algorithm from scratch.
2. A short 2–5-minute video of the task being performed by your agent.

- The video length must **not** exceed 5 minutes (you can speed up the original video to meet this requirement).
- Please zip everything together and submit in one zipped file.

Please do not try to submit the VM because the file size will be too large.

The title of your zip file should include your student ID number and should be entered as the ‘submission title’ when you upload your notebook and video to the submission inbox in Minerva.

You can find the submission area in the submit my work area.

Marking

The breakdown of marks for each section is shown in brackets. Please use these section titles in your technical report. There is a total of 100 marks.

1. Summary and Introduction (3)

2. Creating a suitable Env class that communicates with the provided environment and isolates the ROS requirements from the RL solution (10)
3. Robot model 1: Tabular Solution
    1. Use a tabular RL method (2)
    2. Correctly divide the environment into a grid and use odometry only for state representation (2)
    3. Use a suitable reward function (2)
    4. Parameters tuning and justification (2)
    5. Learning plots, model's success rate (2)
    6. Analysis of the results (2)
4. Robot model 2: Linear function approximation solution
    1. Use a suitable representation for the states (5)
    2. Use a vectorised RL method with linear function approximation (5)
    3. Correctly uses odometry and laser scans for state representation (5)
    4. Use a suitable reward function (5)
    5. Parameters tuning and justification (5)
    6. Learning plots, model's success rate (5)
    7. Analysis of the results (5)
5. Robot model 3: non-linear function approximation solution
    1. Use an off-policy RL method with non-linear function approximation (5)
    2. Use a suitable neural network architecture with an experience-replay buffer (5)
    3. Correctly uses odometry and laser scans as input (5)
    4. Use a suitable reward function (5)

© University of Leeds

5. Parameters tuning and justification (5)
6. Learning plots, model's success rate (5)
7. Analysis of the results (5)

6. Conclusion and Reflection on what went wrong and what worked well (5)

Total marks (100)

This is a compulsory assessment for OCOM5205M Robotics. It is worth 70% of your overall module grade.

You will be marked based on the quality of the technical report.

Your technical report should be around 2000 words in length and will be marked out of a total of 100 marks, according to the mark scheme given in the Jupyter notebook.

Feedback

You will receive your mark and feedback via Minerva, in the **Gradebook** section, accessed from the taps at the top of the page. We will notify you by email when your feedback is available. You will usually receive feedback within 15 working days of the assignment deadline.

Reading List

The reading list and online course readings for this module can be found in the 'Reading List' section of Minerva.

Additional Information and Support

Key Documents

Please ensure that you read and familiarise yourself with the following important documents, available in the Sustainable Business Leadership Online organisation area on Minerva:

- Artificial Intelligence Programme Structure
- Digital Education Service (DES) Code of Practice on Assessment
- Code of Practice on Attendance

Skills@Library

Skills@Library provides support for students to develop academic skills. This includes areas such as researching, writing, note making and referencing. This support is delivered through online resources, workshops and webinars: http://library.leeds.ac.uk/skills.

General Module Forum

The Digital Education Service and the University of Leeds are committed to supporting students. If you have any questions about the academic content of the module, please post them in the module's 'General Module Forum' discussion board. You can access this through the 'Discussion Board' section of Minerva. The Module Team will reply to your post and your peers will also have access to the answer.

Student Success Advisor

If you have any questions about using Minerva, administrative processes, or pastoral care please contact your Student Success Advisor. They will be able to advise you or signpost you to the correct resource.

Recording Policy

As part of your study on this module, webinars will be recorded by the University. This recording is conducted in accordance with the University's Policy on Audio or Video Recording for Educational Purposes. Please read the Policy for further background information relating to

your rights. The purpose of the recording is to support your study at the University. It will be accessible by students enrolled on the module through Minerva.

In accordance with the University's Intellectual Property Policy in general where staff or students make a recording, all rights in that recording (including rights in the sounds made) will belong to the University. The University seeks to respect student and staff rights to be acknowledged as authors and performers. Otherwise, to the extent allowed under the general law, any individual interests in the recordings are waived, allowing the University to act in effect as a custodian, to maximise educational objectives in the general interest of all.

If you have any concerns relating to the recording please contact the individual leading the educational activity, or the module leader. If you wish to opt out of being recorded or wish to request that your contribution be edited please contact the individual leading the activity or the module leader in advance of the session (for optouts) or immediately after the session (for requests for edits). Please note that you may not opt out of recordings which form part of the assessment process on the module. If you have any concerns relation to the recording, please contact your Student Success Advisor.

Complaints

If you have any complaints about this module, please write to DESStudents@leeds.ac.uk and the Digital Education Service will review your complaint. If a remedy cannot be arrived at, the University's official complaints procedure could be followed.